Orchestration for efficient LLM Checkpointing within HPC

Advanced Computing Project 25/26

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Context

LLM training spans several weeks, requiring periodic checkpointing to a Parallel File System (PFS) to prevent total state loss during node failures.

However, checkpointing directly to the PFS creates two problems:

- Simultaneous writes create heavy I/O contention, causing the "noisy neighbor" problem for other jobs.
- High write latency stalls training, leading to significant compute time being wasted.

Problem

Challenge	Impact
PFS Contention	I/O bottlenecks slow training as nodes compete for bandwidth.
Resilience	Hardware failures at scale risk losing weeks of training progress.
Checkpointing	Larger models $\rightarrow$ larger checkpoints $\rightarrow$ higher storage costs/time.
Storage Costs	Frequent PFS writes increase costs and operational overhead.

Table 1: Challenges of LLM Checkpointing to Parallel File Systems

Motivation

The goal is to design a checkpointing system that:

- Mitigates I/O contention on the PFS, ensuring fair resource sharing and Cluster Harmony.
- Provides resilience against hardware failures, minimizing training disruptions.
- Optimizes checkpointing frequency and storage strategies to balance performance and costs.

Approach: Policy-Driven Checkpointing

Controlled PFS Access

Mitigate I/O contention via a centralized **Orchestrator**.

Policy-Driven

Dynamic migration based on cluster-wide scheduling policies.

Traffic Shaping

Token Bucket to smooth traffic and eliminate noisy neighbors.

Tiered Persistence

Local SSDs as a buffer, decoupling training from PFS latency.

Architecture

- TODO

Results

- TODO

Next Steps

- TODO